

Predicting LLM Pretraining Loss Curves Across Learning-Rate Schedules

Revisiting the capacity of MTL through a broader parameter search

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Roadmap

The presentation follows the scientific argument in four stages.

- ① We formulate schedule-to-loss prediction and introduce three published baselines: MTL, MPL, and FSL.
- ② We develop **Tuned-MTL**, our main contribution, by expanding the memory-scale search and revising the fitting emphasis.
- ③ We evaluate all methods under a common cosine-to-WSD protocol and interpret their prediction curves.
- ④ We summarize several smaller exploratory variants and the broader lessons from the comparison.

1 Problem Formulation

2 Published Baselines

3 Main Contribution: Tuned-MTL

4 Experimental Design

5 Quantitative Results

6 Qualitative Results

7 Takeaways

Why Predict an Entire Loss Curve?

A full pretraining run is expensive, while schedule decisions must often be made before that run is available.

- A loss-curve model estimates training progress under a candidate learning-rate schedule.
- Curve prediction reveals when two schedules begin to diverge, rather than reporting only their final losses.
- A transferable model can support schedule design using a small number of observed training curves.

The central difficulty is historical dependence: the loss at step t reflects the complete optimization path up to that step.

Schedule-to-Loss Prediction

For a training horizon T , let

$$\boldsymbol{\eta} = (\eta_1, \eta_2, \dots, \eta_T)^\top$$

denote the learning-rate schedule. We model the loss at step t as a functional of the schedule prefix:

$$L_t = L_\Theta(\eta_{\leq t}), \quad \eta_{\leq t} = (\eta_1, \eta_2, \dots, \eta_t).$$

The parameters Θ are fitted from an observed curve. Transfer requires the same fitted functional to predict a curve produced by a different schedule.

Course Task and Research Question

The main experiment fits each method on the loss curve produced by a cosine schedule and evaluates its prediction on a warmup-stable-decay (WSD) schedule.

cosine curve for fitting → WSD curve for evaluation.

A three-stage 8-1-1 schedule provides a second, auxiliary transfer case.

Research question. How much of the observed transfer performance comes from adding model complexity, and how much remains available inside the parameter space of the earliest MTL baseline?

Three Baseline Perspectives

The three published methods encode schedule history in different ways.

Method	Central object	Intuition
MTL	exponentially decaying memory	summarize learning-rate changes with one persistent state
MPL	a sum of power-law responses	treat each learning-rate decrease as a source with its own delayed effect
FSL	intrinsic time and convolution	measure progress by accumulated learning rate and convolve past schedule changes

These perspectives provide increasingly expressive descriptions of schedule response, with correspondingly different data requirements.

Baseline 1: Momentum Law (MTL)

MTL models the predicted loss as

$$\hat{L}_t = L_0 + AT_t^{-\alpha} - CS_2(t), \quad T_t = \epsilon + \sum_{i \leq t} \eta_i.$$

- $L_0 + AT_t^{-\alpha}$ is a power law in accumulated learning rate, which acts as a measure of training progress.
- $CS_2(t)$ represents the additional loss reduction associated with learning-rate annealing.
- The model uses four continuous parameters and one memory parameter, λ .

Tissue, Wang, and Wang, *Scaling Law with Learning Rate Annealing*, arXiv:2408.11029, 2024.

MTL Memory and Parameter Meanings

The annealing state obeys

$$m_t = \lambda m_{t-1} + (\eta_{t-1} - \eta_t), \quad S_2(t) = \sum_{i \leq t} m_i.$$

A learning-rate decrease creates a positive impulse. The factor λ controls how slowly that impulse fades.

Parameter	Interpretation
L_0	asymptotic loss floor
A, α	scale and exponent of the progress term
C	magnitude of the annealing contribution
λ	persistence of schedule memory; values near one retain longer history

Tissue, Wang, and Wang, arXiv:2408.11029, 2024.

Baseline 2: Multi-Power Law (MPL)

MPL associates every positive learning-rate decrease

$$\Delta_i^+ = \max(\eta_{i-1} - \eta_i, 0)$$

with a delayed power-law response:

$$\hat{L}_t = L_0 + AT_t^{-\alpha} - B \sum_{i \leq t} \Delta_i^+ \left[1 - \left(1 + C\eta_i^{-\gamma}(T_t - T_i + \eta_i) \right)^{-\beta} \right].$$

The total schedule effect is the sum of these responses. MPL can therefore represent different effects for the timing and magnitude of multiple decay events.

Luo et al., *A Multi-Power Law for Loss Curve Prediction Across Learning Rate Schedules*, arXiv:2503.12811, 2025.

How to Read the MPL Parameters

MPL retains the progress term from MTL and adds a flexible response kernel.

- L_0 , A , and α describe the loss floor and the power-law improvement with accumulated learning rate.
- B scales the total benefit assigned to learning-rate decreases.
- C sets the response timescale, while β controls the response shape.
- γ changes the response according to the learning rate at which a decrease occurs.

The additional flexibility can distinguish several decay events, although estimating seven coupled parameters from one fitting curve is statistically demanding.

Luo et al., arXiv:2503.12811, 2025.

Baseline 3: Functional Scaling Laws (FSL)

Functional Scaling Laws define training progress through intrinsic time:

$$T(k) = \sum_{i \leq k} \eta_i.$$

Two training steps with different learning rates therefore represent different amounts of progress. The practical loss model has the form

$$\hat{L}_k = L_0 + \frac{c_1}{T(k)^s} - \text{LRD}(k),$$

where $\text{LRD}(k)$ is a convolutional response to earlier learning-rate decreases.

Li, Chen, Huang, Wang, and Wu, *Functional Scaling Laws in Kernel Regression*, arXiv:2509.19189, 2025.

FSL Schedule-Response Kernel

For fixed model and data size, the practical response used in our reproduction is

$$\text{LRD}(k) = c_2 \sum_{i \leq k} \Delta_i^+ (c_3 + T(i)^{-s}) [1 - (1 + c_4(T(k) - T(i)))^{-\gamma}] .$$

- c_1 and s determine the progress law.
- c_2 scales the combined schedule response, and c_3 adjusts source strength.
- c_4 and γ control how the effect of each decay event grows with intrinsic-time separation.

Li et al., arXiv:2509.19189, 2025; practical LLM ansatz in Appendix B.2.

Baseline Comparison Before Tuning

Method	Schedule memory	Continuous parameters	Main inductive bias
MTL	one exponential scale	four, plus λ	persistent state
MPL	event-wise power response	seven	decay superposition
FSL	intrinsic-time convolution	seven	convolutional kernel

MTL is the oldest and lowest-dimensional method in this comparison. Its compact structure becomes valuable when only one curve is available for fitting.

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Central Finding: MTL Has Substantial Unused Capacity

The clean baseline comparison identifies MTL as the strongest of the three published methods under the one-curve fitting regime.

- The default MTL search examines only five values of λ .
- Near $\lambda = 1$, a small numerical change produces a large change in memory duration.
- A sparse grid can therefore exclude the timescale needed for transfer to WSD.

Our main contribution is **Tuned-MTL**: an estimation procedure that searches a broader memory space and places greater fitting emphasis on the informative late portion of the cosine curve.

Tuned-MTL Retains the Complete MTL Model

Tuned-MTL uses the same predictive equation:

$$\hat{L}_t = L_0 + A \left(\epsilon + \sum_{i \leq t} \eta_i \right)^{-\alpha} - C \sum_{i \leq t} m_i, \quad m_i = \lambda m_{i-1} + (\eta_{i-1} - \eta_i).$$

The estimation procedure changes in three places:

- ① the λ grid expands from five values to thirteen values concentrated near one;
- ② selected points in the final 20% of the cosine curve receive weight 3, compared with weight 2 in the baseline fit;
- ③ fifty random restarts explore the expanded grid, compared with twenty for the baseline.

Why the Memory Search Matters

Exponential memory has an approximate half-life

$$h_{1/2} = \frac{\log(1/2)}{\log \lambda}.$$

Fit	Selected λ	Approximate half-life
Baseline MTL	0.99	69 recorded steps
Tuned-MTL	0.9997	2310 recorded steps

Tuned-MTL selects a much longer schedule memory. This allows an annealing event to influence predictions over a substantial fraction of training.

What Tail Weighting Changes

Let

$$r_t = \log \hat{L}_t - \log \bar{L}_t$$

be the residual against the smoothed cosine loss. Tuned-MTL minimizes

$$\sum_{t \in I_{\text{fit}}} w_t H_{\delta}(r_t), \quad w_t = \begin{cases} 3, & t \text{ lies in the final 20\% of the fitting region,} \\ 1, & \text{otherwise.} \end{cases}$$

Per selected point, late cosine observations contribute three times as strongly. Empirically, this emphasis helps identify the long-memory solution that transfers accurately across the full WSD curve.

Selected Tuned-MTL Parameters

Parameter	Fitted value	Role in the model
L_0	2.72857	asymptotic loss floor
A	774.097	scale of the progress term
α	0.93428	progress-law exponent
C	4.936×10^{-5}	annealing-response magnitude
λ	0.9997	memory persistence

The numerical scale of A and C follows the normalization of the learning-rate schedule. Their functional roles are given by the complete equation on the preceding slides.

Secondary Explorations

After establishing Tuned-MTL, we tested several smaller modifications.

- **Two-scale and three-scale MTL** mix fixed exponential memories to approximate multiple timescales.
- **Compact FSL-MPL** replaces the MPL kernel with a simpler intrinsic-time response.
- **Source-weighted FSL-MPL** lets the strength of a decay event depend on when it occurs.
- **Residual-corrected FSL-MPL** fits a regularized correction to the cosine residuals.

These variants are exploratory contributions. They probe alternative inductive biases and remain secondary to Tuned-MTL in the final comparison.

Lesson from the Exploratory Variants

The exploratory models add expressive schedule-response mechanisms, yet none reaches the full-curve transfer accuracy of Tuned-MTL.

- Multiple memory scales are difficult to identify from a single cosine trajectory.
- Event-wise kernels can improve selected regions while introducing bias elsewhere.
- Residual correction improves average behavior in the stable region and loses calibration during WSD decay.

The comparison favors a compact model whose relevant timescale is explicitly included in the search space.

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Course Data

The dataset contains three loss curves with the same model size and data budget; only the learning-rate schedule changes.

Schedule	Observations	Final step	Experimental role
Cosine	33,907	33,906	parameter fitting
WSD	33,907	33,906	main evaluation
8-1-1	33,908	33,907	auxiliary evaluation

In the 8-1-1 schedule, the learning rate changes at 80% and 90% of training, producing three levels:

$$\eta, \quad \eta/\sqrt{10}, \quad \eta/10.$$

Learning-Rate Schedules



Cosine decays continuously; WSD has a stable phase; 8-1-1 introduces two discrete reductions.

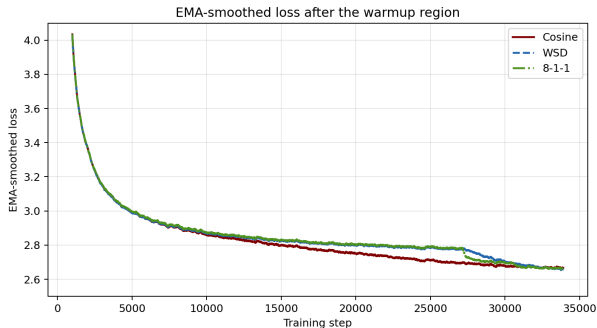
Observation and Prediction Target

The recorded loss contains high-frequency training noise. We form an exponentially weighted moving average:

$$\bar{L}_t = (1 - a)\bar{L}_{t-1} + aL_t, \quad a = \frac{2}{201 + 1}.$$

- The span parameter 201 determines the geometric rate at which older observations lose influence.
- The resulting curve preserves schedule-scale trends while reducing local noise.
- All headline metrics compare predictions with this smoothed target.

Observed Loss Curves After Warmup



The plot begins at step 1000 and caps the vertical axis at 4.1 to expose schedule differences.

Fitting and Evaluation Protocol

Every method follows the same main experimental sequence.

- ① We fit model parameters to the smoothed cosine curve from step 1000 onward.
- ② The fitting set uses one observation every 800 steps, with denser sampling every 250 steps in the final 20% of the fitting region.
- ③ We evaluate the fitted model on WSD every 20 steps and include the final observation.
- ④ We apply the same evaluation to 8-1-1 as auxiliary evidence.

WSD and 8-1-1 values do not enter the continuous parameter fitting objective.

Huber Loss on Logarithmic Residuals

The fitting residual is

$$r_t = \log \hat{L}_t - \log \bar{L}_t.$$

We apply the Huber function

$$H_\delta(r) = \begin{cases} \frac{1}{2}r^2, & |r| \leq \delta, \\ \delta (|r| - \frac{1}{2}\delta), & |r| > \delta, \end{cases} \quad \delta = 10^{-3}.$$

Logarithmic residuals measure proportional disagreement, while the linear Huber regime limits the influence of occasional large deviations.

How the Reported Metrics Are Calculated

For observed values $y_t = \bar{L}_t$, predictions $\hat{y}_t$, and the WSD evaluation index set I :

$$\text{RMSE} = \sqrt{\frac{1}{|I|} \sum_{t \in I} (\hat{y}_t - y_t)^2}, \quad \text{MAE} = \frac{1}{|I|} \sum_{t \in I} |\hat{y}_t - y_t|,$$

$$R^2 = 1 - \frac{\sum_{t \in I} (y_t - \hat{y}_t)^2}{\sum_{t \in I} (y_t - \bar{y})^2}, \quad \text{FinalE} = |\hat{y}_{t_{\text{final}}} - y_{t_{\text{final}}}|.$$

Lower values are better for RMSE, MAE, and FinalE; higher values are better for R^2 .

Common Implementation and Evaluation

We adapted the released method implementations to the course data and added an adapter that reads the provided training step, loss, and learning-rate observations directly.

- Each baseline retains its published mathematical formulation.
- A shared evaluation layer applies the same smoothing, evaluation indices, and metric definitions.
- Formula-level checks compare adapted evaluators with the corresponding reference computations under identical parameters.
- Fixed random seeds and restart counts control optimizer variation across repeated runs.

Implementation commands, artifact schemas, dependency versions, and integrity checks are documented in the repository.

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Primary Finding

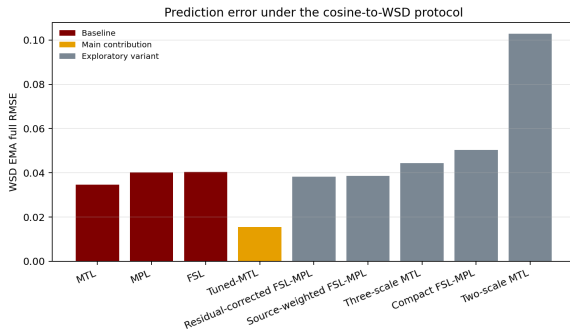
Our primary finding is that Tuned-MTL achieves a full-region WSD RMSE of 0.015461, compared with 0.034569 for baseline MTL. This corresponds to a 55.3% reduction in RMSE and reveals substantial performance headroom within the original MTL model family.

WSD Results on the Smoothed Full Curve

Method	RMSE ↓	MAE ↓	R^2 ↑	FinalE ↓
MTL	<i>0.034569</i>	0.028816	<i>0.963640</i>	0.004317
MPL	0.040219	0.031148	0.950783	0.029069
FSL	0.040358	0.035311	0.950443	0.053288
Tuned-MTL	0.015461	0.010069	0.992727	0.016741
Residual-corrected FSL-MPL	0.038326	0.027214	0.955309	0.081484
Source-weighted FSL-MPL	0.038635	<i>0.026214</i>	0.954584	0.027232
Three-scale MTL	0.044395	0.035417	0.940033	0.064200
Compact FSL-MPL	0.050309	0.037628	0.922993	<i>0.011667</i>
Two-scale MTL	0.102856	0.090321	0.678114	0.050873

Values compare predictions with the EMA-smoothed WSD curve from step 1000, sampled every 20 steps. Bold marks the best value and italics mark the second best.

Method Groups in the Main Comparison



Published baselines appear first, followed by Tuned-MTL and the exploratory variants.

Where Tuned-MTL Gains Accuracy

WSD region	Baseline MTL RMSE	Tuned-MTL RMSE
Full evaluation curve	<i>0.034569</i>	0.015461
Stable phase	<i>0.037392</i>	0.007664
Final 20%	0.018448	<i>0.030976</i>
Final 1000 steps	0.004565	<i>0.020278</i>

Tuned-MTL's full-curve advantage comes primarily from the long stable phase. Baseline MTL remains more accurate near the endpoint, so the preferred setting depends on the evaluation objective.

Direct Comparison with Baseline MTL

Metric	Baseline MTL	Tuned-MTL
WSD RMSE	<i>0.034569</i>	0.015461
WSD MAE	<i>0.028816</i>	0.010069
WSD MAPE	<i>0.010213</i>	0.003591
WSD R^2	<i>0.963640</i>	0.992727
WSD FinalE	0.004317	<i>0.016741</i>
WSD WorstE	0.063370	<i>0.082189</i>

Tuned-MTL gives the most accurate global trajectory. Baseline MTL provides tighter endpoint and worst-case calibration.

Auxiliary Transfer to the 8-1-1 Schedule

Method	RMSE ↓	MAE ↓	R^2 ↑
MTL	<i>0.030190</i>	<i>0.024659</i>	<i>0.973120</i>
MPL	0.041732	0.030597	0.948638
FSL	0.041044	0.033478	0.950317
Tuned-MTL	0.013594	0.008491	0.994550

- Tuned-MTL also attains the lowest error on this second schedule.
- Baseline MTL remains the strongest published baseline.
- The 8-1-1 curve is evaluated with the same fitted parameters and metric definitions.

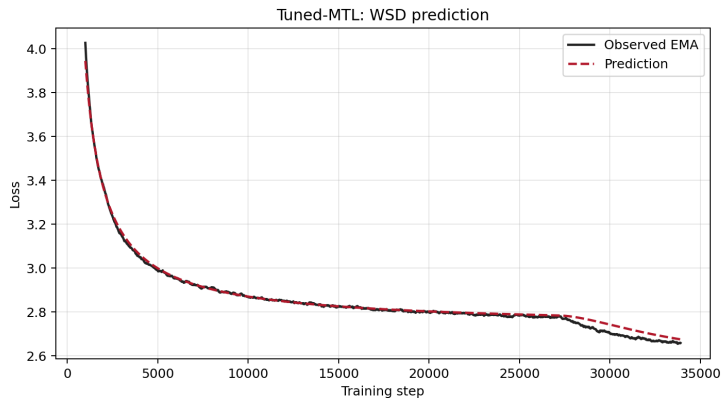
Quantitative Interpretation

The results support three conclusions.

- A low-dimensional schedule model is well matched to the one-curve fitting regime.
- Memory-scale coverage is decisive: $\lambda = 0.9997$ makes much longer history available to MTL.
- Full-curve error and endpoint error reward different calibration choices.
Tuned-MTL is strongest for the declared full-region objective, while baseline MTL remains stronger at the final point.

The auxiliary schedule follows the same overall ranking and strengthens the evidence for the long-memory setting.

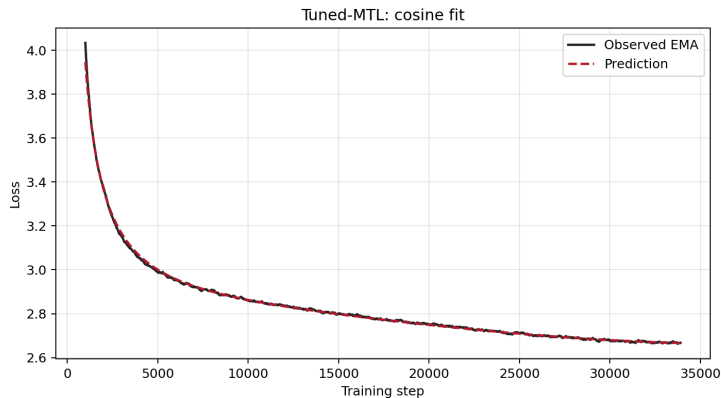
Tuned-MTL WSD Prediction



Tuned-MTL follows the stable WSD phase closely, where its RMSE is 0.007664.

During final decay, the prediction stays above the observed curve. This produces the larger tail and endpoint errors seen in the regional table.

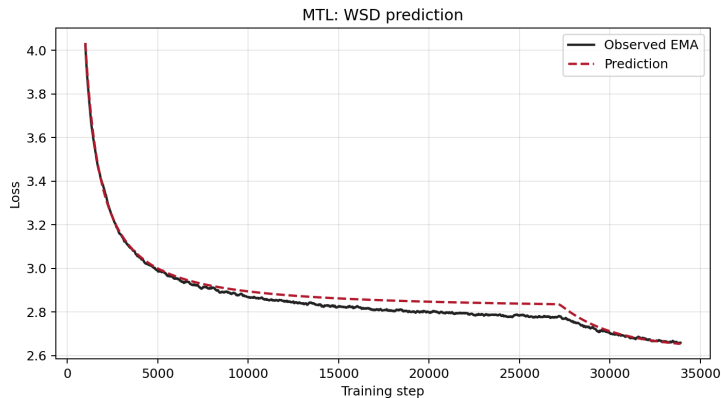
Tuned-MTL Cosine Fit and Transfer



The fitted cosine curve remains smooth across the full training horizon.

Tail weighting keeps the fit sensitive to the cosine decay region, while the selected λ supplies the long memory needed for transfer.

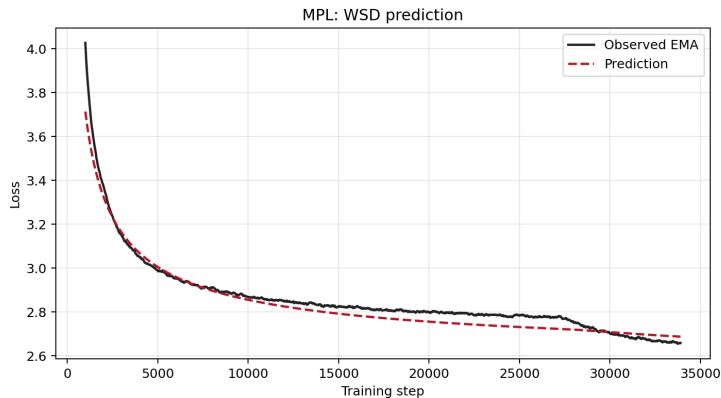
Baseline MTL WSD Prediction



Baseline MTL captures the overall WSD shape.

Its positive bias through much of the stable phase raises full-curve RMSE. Near the endpoint, the prediction converges closely to the observed loss, giving the best FinalE.

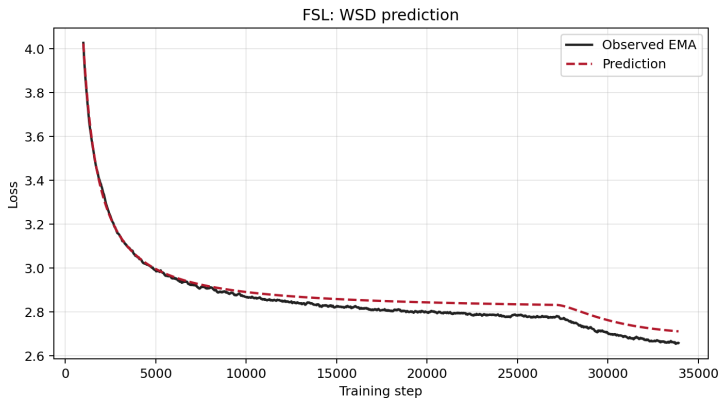
Baseline MPL WSD Prediction



MPL adapts visibly to the WSD decay transition.

A large early deviation and a negative bias through the stable phase dominate its full error. The final prediction then moves above the observed curve.

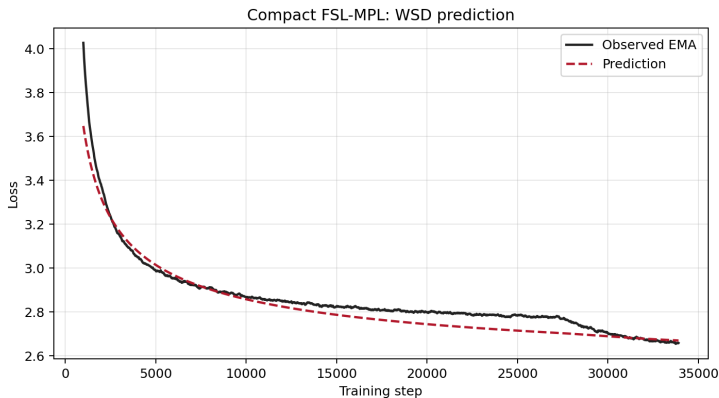
Baseline FSL WSD Prediction



FSL gives a smooth intrinsic-time prediction.

It develops a systematic positive bias during WSD decay: the mean decay-region error is 0.0574. This bias explains its larger tail and final errors.

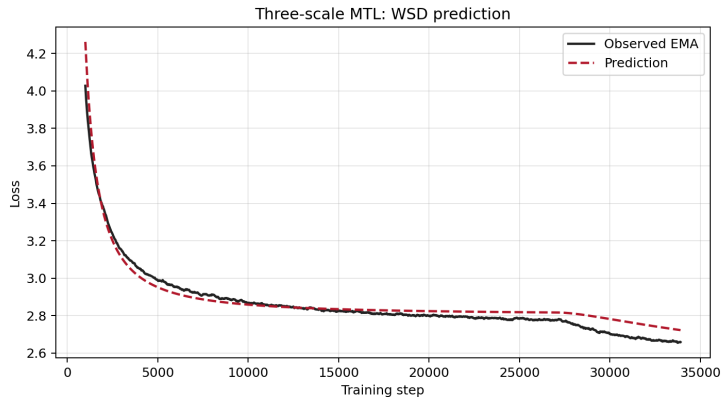
Auxiliary Figure: Compact FSL-MPL



This exploratory variant underpredicts much of the stable phase.

Its endpoint calibration is comparatively strong, although its full-curve RMSE remains above all three published baselines.

Auxiliary Figure: Three-Scale MTL



The three-scale mixture follows the stable region reasonably well.

It retains too much positive correction through WSD decay, producing a tail RMSE of 0.0713. One cosine curve provides limited information for estimating several memory weights.

Qualitative Synthesis

The prediction curves clarify the aggregate metrics.

- Tuned-MTL's long memory corrects the broad stable-phase bias that dominates baseline MTL's RMSE.
- Baseline MTL forgets schedule changes more quickly and aligns more closely with the final WSD observations.
- MPL and FSL respond to decay through richer kernels, yet their fitted responses introduce systematic regional biases.
- The auxiliary variants distribute error differently without improving the main full-curve objective.

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Takeaways

- **Tuned-MTL is the main contribution.** A broader memory search and stronger late-curve emphasis reduce WSD full-region RMSE by 55.3% relative to baseline MTL.
- The selected $\lambda = 0.9997$ corresponds to a memory half-life of roughly 2310 recorded steps, revealing substantial capacity inside the earliest baseline family.
- MPL, FSL, and the exploratory variants provide useful alternative inductive biases, while their additional parameters are harder to estimate from one fitting curve.
- Metric choice matters: Tuned-MTL leads on the global trajectory, and baseline MTL leads on endpoint calibration.

Scope and Limitations

- The course dataset contains three curves at one model size and one data budget.
- Only the cosine curve is available for continuous parameter fitting, which limits identification of flexible response kernels.
- The quantitative conclusions apply to the declared EMA-smoothed, post-warmup, full-region objective.
- Broader validation should include additional model sizes, training horizons, and independently generated schedules.

Project Repository

We present our project repository at

https://github.com/VincentWangzt/TDLT_project.

The repository provides:

- the shared data adapter and published baseline implementations;
- the Tuned-MTL configuration and the exploratory variants;
- fixed-seed experiment entry points and generated prediction tables;
- metric definitions, environment specifications, tests, and documentation.

The README records the complete reproduction procedure and implementation details.

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8 Appendix

Appendix: MTL Memory Grids

The baseline search uses

$$\{0.95, 0.99, 0.995, 0.999, 0.9995\}.$$

Tuned-MTL searches

$$\{0.9, 0.95, 0.97, 0.99, 0.993, 0.995, 0.997, \\ 0.999, 0.9993, 0.9995, 0.9997, 0.9999, 0.99995\}.$$

The expanded grid resolves long memory scales much more finely near one. The selected value is $\lambda = 0.9997$.

Appendix: Additional Error Definitions

For errors $e_t = \hat{y}_t - y_t$:

$$\text{MAPE} = \frac{1}{|I|} \sum_{t \in I} \frac{|e_t|}{|y_t|}, \quad \text{WorstE} = \max_{t \in I} |e_t|.$$

Regional RMSE uses the same definition as full RMSE after restricting I :

- **Stable phase**: observations before the WSD learning rate begins to decay.
- **Final 20%**: the last fifth of the evaluation interval.
- **Final 1000 steps**: observations within 1000 steps of the final point.

Appendix: References

- ① Howe Tissue, Venus Wang, and Lu Wang. *Scaling Law with Learning Rate Annealing*. 2024. arXiv:2408.11029.
- ② Kairong Luo, Haodong Wen, Shengding Hu, Zhenbo Sun, Zhiyuan Liu, Maosong Sun, Kaifeng Lyu, and Wenguang Chen. *A Multi-Power Law for Loss Curve Prediction Across Learning Rate Schedules*. 2025. arXiv:2503.12811.
- ③ Binghui Li, Fengling Chen, Zixun Huang, Lean Wang, and Lei Wu. *Functional Scaling Laws in Kernel Regression: Loss Dynamics and Learning Rate Schedules*. 2025. arXiv:2509.19189.
- ④ Huaqing Zhang, Kaiyue Wen, and Tengyu Ma. *Configuration-to-Performance Scaling Law with Neural Ansatz*. 2026. arXiv:2602.10300.

References 1–3 support the reproduced baseline formulations. Reference 4 motivates the small residual-correction exploration.